INFIX TO POSTFIX

Specs.

- Your project includes two parts:
  - First, you need to implement an algorithm for **infix to postfix conversion** using stacks.
  - Second, implement an algorithm to **evaluate the postfix expression** your code outputs.

- Write **your own solution** in **C/C++**
- Due Date: **13th week.**
- **Individual** or **team (max 2)**
- Out of (**15 marks**)
- **Submit Code and Report (Technical Report)**

Infix to Postfix Conversion

- Initialize a stack of **characters** to hold the operators (^, *, /, +, -)
- Parse the infix string left to right:
  - If an **operand** is read:
    - Then output it into the postfix expression
  - Else if **'('** is read:
    - Then push it into the stack
  - Else if **')'** is read:
    - Then pop and output into the postfix expression until the matching '(' is popped from the stack
  - Else if an **operator X** is read:
    - If the **stack is empty** or character **Y (top of the stack)** = '(' or **Y** is an operator of **lower precedence** than **X**, then push **X**
    - While **Y** is an operator of **higher or equal precedence** than **X**, pop and output **Y**
    - Then push **X**
- When we reach the end of the input, pop and output until the stack is empty

Postfix Expression Evaluation

- Initialize a stack to hold **integer values** (operands)
- Parse the postfix expression left to right:
  - If an **operand** is read:
    - Push it into the stack
  - Else if an **operator** is read:
    - Pop the **first two elements** on top of the stack and **do the operation** on them, then **push the result** back into the stack
- When we reach the end of the postfix expression, **output the top of the stack** which is the result of the whole expression.

Example – I

- Infix: 3 + 4 * 5 - 2

Example – I (Conversion)

- Infix: **3** + 4 * 5 - 2
- Stack:
- Postfix:

Example – I (Conversion)

- Infix: 3 + 4 * 5 - 2
- Stack:
- Postfix: 3

Example – I (Conversion)

- Infix: 3 + **4** * 5 - 2
- Stack: +
- Postfix: 3

Example – I (Conversion)

- Infix: 3 + 4 * 5 - 2
- Stack: +
- Postfix: 3 4

Example – I (Conversion)

- Infix: 3 + 4 * **5** - 2
- Stack: + *
- Postfix: 3 4

Example – I (Conversion)

- Infix: 3 + 4 * 5 - 2
- Stack: + *
- Postfix: 3 4 5

Example – I (Conversion)

- Infix: 3 + 4 * 5 - 2
- Stack: +
- Postfix: 3 4 5 *

Example – I (Conversion)

- Infix: 3 + 4 * 5 - **2**
- Stack: -
- Postfix: 3 4 5 * +

Example – I (Conversion)

- Infix: 3 + 4 * 5 - 2
- Stack: -
- Postfix: 3 4 5 * + 2

Example – I (Conversion)

- Infix: 3 + 4 * 5 - 2
- Stack:
- Postfix: **3 4 5 * + 2 -**

- **DONE!**

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -

Example – I (Evaluation)

- Postfix: **3** 4 5 * + 2 -
- Stack:

Example – I (Evaluation)

- Postfix: 3 **4** 5 * + 2 -
- Stack: 3

Example – I (Evaluation)

- Postfix: 3 4 **5** * + 2 -
- Stack: 3 4

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack: 3 4 5

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack: 3 4

- operand1 = 5

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack: 3

- operand1 = 5
- operand2 = 4

Example – I (Evaluation)

- Postfix: 3 4 5 ***** + 2 -
- Stack: 3

- operand1 = 5
- operand2 = 4
- operand2 ***** operand1 = 4 * 5 = **20**

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack: 3 20

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack: 3

- operand1 = 20

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack:

- operand1 = 20
- operand2 = 3

Example – I (Evaluation)

- Postfix: 3 4 5 * **+** 2 -
- Stack:

- operand1 = 20
- operand2 = 3
- operand2 **+** operand1 = 3 + 20 = **23**

Example – I (Evaluation)

- Postfix: 3 4 5 * + **2** -
- Stack: 23

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack: 23 2

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack: 23

- operand1 = 2

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack:

- operand1 = 2
- operand2 = 23

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 **-**
- Stack:

- operand1 = 2
- operand2 = 23
- operand2 **-** operand1 = 23 - 2 = **21**

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack: 21

Example – I (Evaluation)

- Postfix: 3 4 5 * + 2 -
- Stack:

- Output: **21**

- **DONE!**

Example – II

- Infix: (300 + 20 / 5) * (43 - 21)

Example – II (Conversion)

- Infix: **(** 300 + 20 / 5) * (43 - 21)
- Stack:
- Postfix:

Example – II (Conversion)

- Infix: (**300** + 20 / 5) * (43 - 21)
- Stack: (
- Postfix:

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack: (
- Postfix: 300

Example – II (Conversion)

- Infix: (300 + **20** / 5) * (43 - 21)
- Stack: (+
- Postfix: 300

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack: (+
- Postfix: 300 20

Example – II (Conversion)

- Infix: (300 + 20 / **5**) * (43 - 21)
- Stack: (+ /
- Postfix: 300 20

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack: (+ /
- Postfix: 300 20 5

Example – II (Conversion)

- Infix: (300 + 20 / 5 **)** * (43 - 21)
- Stack: (+
- Postfix: 300 20 5 /

Example – II (Conversion)

- Infix: (300 + 20 / 5 **)** * (43 - 21)
- Stack: (
- Postfix: 300 20 5 / +

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack:
- Postfix: 300 20 5 / +

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack: *
- Postfix: 300 20 5 / +

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (**43** - 21)
- Stack: * (
- Postfix: 300 20 5 / +

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack: * (
- Postfix: 300 20 5 / + 43

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - **21**)
- Stack: * (-
- Postfix: 300 20 5 / + 43

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack: * (-
- Postfix: 300 20 5 / + 43 21

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack: * (
- Postfix: 300 20 5 / + 43 21 -

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack: *
- Postfix: 300 20 5 / + 43 21 -

Example – II (Conversion)

- Infix: (300 + 20 / 5) * (43 - 21)
- Stack:
- Postfix: **300 20 5 / + 43 21 - ***

- **DONE!**

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *

Example – II (Evaluation)

- Postfix: **300** 20 5 / + 43 21 - *
- Stack:

Example – II (Evaluation)

- Postfix: 300 **20** 5 / + 43 21 - *
- Stack: 300

Example – II (Evaluation)

- Postfix: 300 20 **5** / + 43 21 - *
- Stack: 300 20

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 300 20 5

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 300 20

- operand1 = 5

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 300

- operand1 = 5
- operand2 = 20

Example – II (Evaluation)

- Postfix: 300 20 5 **/** + 43 21 - *
- Stack: 300

- operand1 = 5
- operand2 = 20
- operand2 **/** operand1 = 20 / 5 = **4**

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 300 4

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 300

- operand1 = 4

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack:

- operand1 = 4
- operand2 = 300

Example – II (Evaluation)

- Postfix: 300 20 5 / **+** 43 21 - *
- Stack:

- operand1 = 4
- operand2 = 300
- operand2 **+** operand1 = 300 + 4 = **304**

Example – II (Evaluation)

- Postfix: 300 20 5 / + **43** 21 - *
- Stack: 304

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 **21** - *
- Stack: 304 43

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 304 43 21

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 304 43

- operand1 = 21

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 304

- operand1 = 21
- operand2 = 43

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 **-** *
- Stack: 304

- operand1 = 21
- operand2 = 43
- operand2 **-** operand1 = 43 - 21 = **22**

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 304 22

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 304

- operand1 = 22

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack:

- operand1 = 22
- operand2 = 304

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *****
- Stack:

- operand1 = 22
- operand2 = 304
- operand2 ***** operand1 = 304 * 22 = **6688**

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack: 6688

Example – II (Evaluation)

- Postfix: 300 20 5 / + 43 21 - *
- Stack:

- Output: **6688**

- **DONE!**

Special Case

- Power associates to the **right** not to the left as other operators. For example:
- Infix: “2 ^ 3 ^ 2”
  - Correct Postfix: 2 3 2 ^ ^
  - Correct Result: 2 ^ 9 = 512

  - Incorrect Postfix: 2 3 ^ 2 ^
  - Incorrect Result: 64